

Lignans in man and in animal species.



In our laboratories, for several years, two phenolic compounds have been detected during gas chromatographic-mass spectrometric analysis of urinary steroid extracts from human and animal species. Although features of the mass spectra of their trimethylsilyl (TMS) ether derivatives resembled those of oestrogens, they were atypical of steroids. The possibility that they were artefacts of the isolation procedures was discounted after careful studies with blanks, by varying the extraction method and because they were present almost exclusively as conjugates of glucuronic acid. Several of the general characteristics of the unknown compounds were reported after one (referred to as compound 180/442) was found to have a cyclic pattern of excretion during the menstrual cycle of an adult vervet monkey (Fig. 1). An investigation of the nature and distribution of the compounds has shown them to be urinary constituents in humans, baboons, vervet monkeys and rats, and further related compounds have been detected, so far only in vervet monkey urine. We now report spectroscopic and chemical studies that show the two original compounds to be lignans, which have a 2,3-dibenzylbutane skeleton as their basic structure. Unlike all previously known natural lignans, invariably of plant origin, the two mammalian compounds carry phenolic hydroxy groups only in the meta position of the aromatic rings.